Calibration results
==========
Normalized Residuals

Reprojection error (cam0): mean 0.23477049618328305, median 0.1872273142280729, std: 0.20348250573464385
Gyroscope error (imu0): mean 0.14879105849977012, median 0.10132046423852148, std: 0.17284386314247058
Accelerometer error (imu0): mean 0.14263271684972015, median 0.09485051860817878, std: 0.19053446207048666

Residuals

Reprojection error (cam0) [px]: mean 0.23477049618328305, median 0.1872273142280729, std: 0.20348250573464385
Gyroscope error (imu0) [rad/s]: mean 0.0117497757388359, median 0.008001103994829162, std: 0.013649184636731456
Accelerometer error (imu0) [m/s^2]: mean 0.11075371276296804, median 0.07365103410613047, std: 0.14794921915310003

Transformation (cam0):

T_ci: (imu0 to cam0):
[[-0.2242458 -0.95212715 0.20776839 -0.03489144]
 [0.38705843 -0.28267787 -0.8776554   0.01810612]
 [0.89437106 -0.11639203 0.43191816 -0.01713891]
 [0.   0.   0.   1.   ]]

T_ic: (cam0 to imu0):
[[-0.2242458   0.38705843 0.89437106 0.00049616]
 [-0.95212715 -0.28267787 -0.11639203 -0.03009772]
 [0.20776839 -0.8776554   0.43191816 0.03054288]
 [0.   0.   0.   1.   ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
0.008782276351565543

Gravity vector in target coords: [m/s^2]
[-0.19002627 -9.40818526 2.76013822]

Calibration configuration
==========

cam0

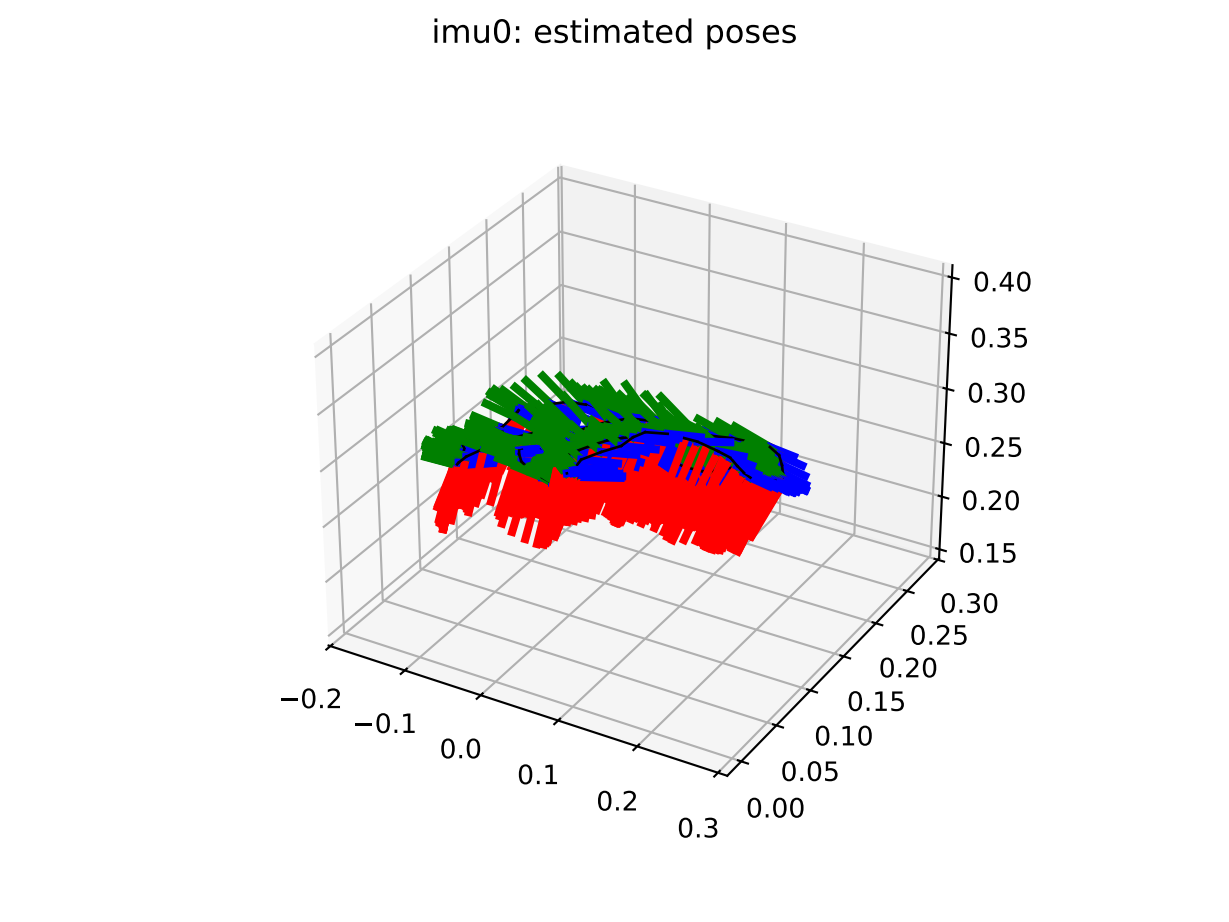
  Camera model: pinhole
  Focal length: [652.879962242561, 652.6913800309958]
  Principal point: [402.7459765146756, 293.51689967360625]
  Distortion model: radtan
  Distortion coefficients: [0.19004630188120233, -0.3566296810567926, -0.001643546078163314,
0.0007208195618292788]
  Type: aprilgrid
  Tags:
    Rows: 6
    Cols: 6
    Size: 0.017733333333333334 [m]
    Spacing 0.00532 [m]

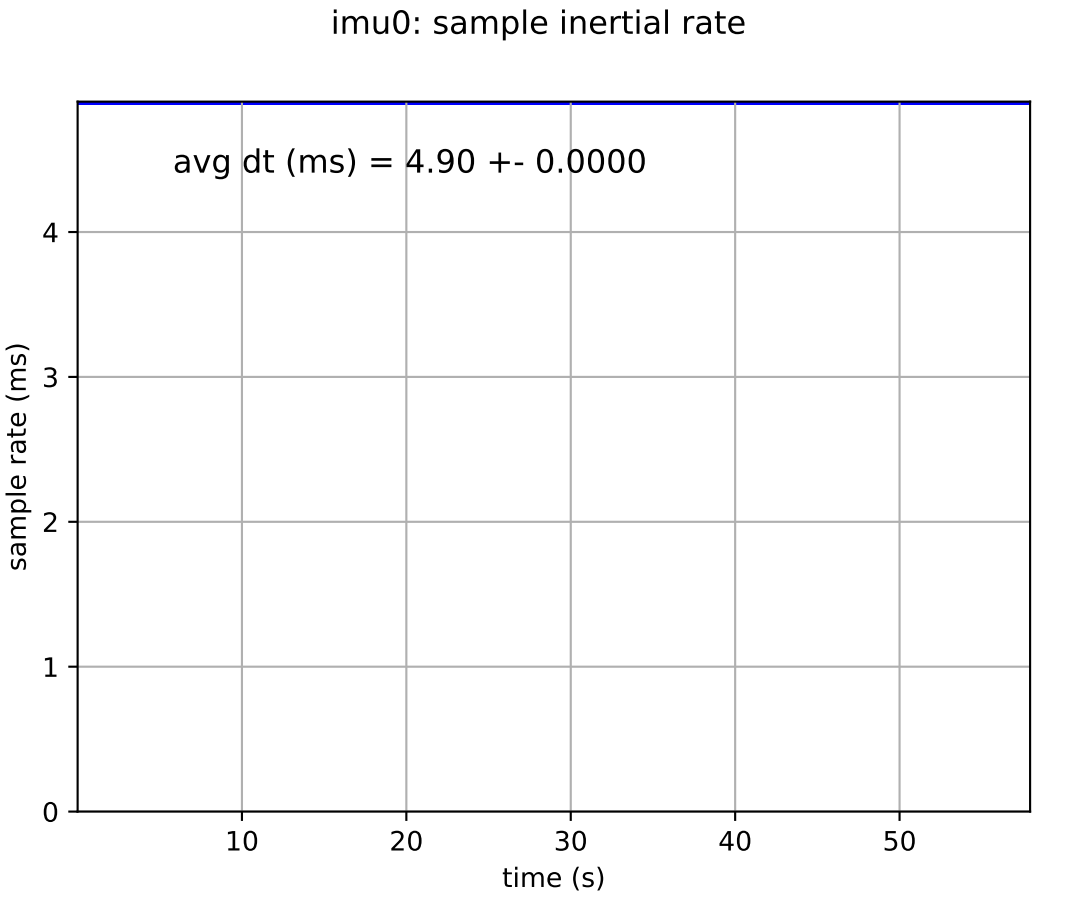
IMU configuration
==========

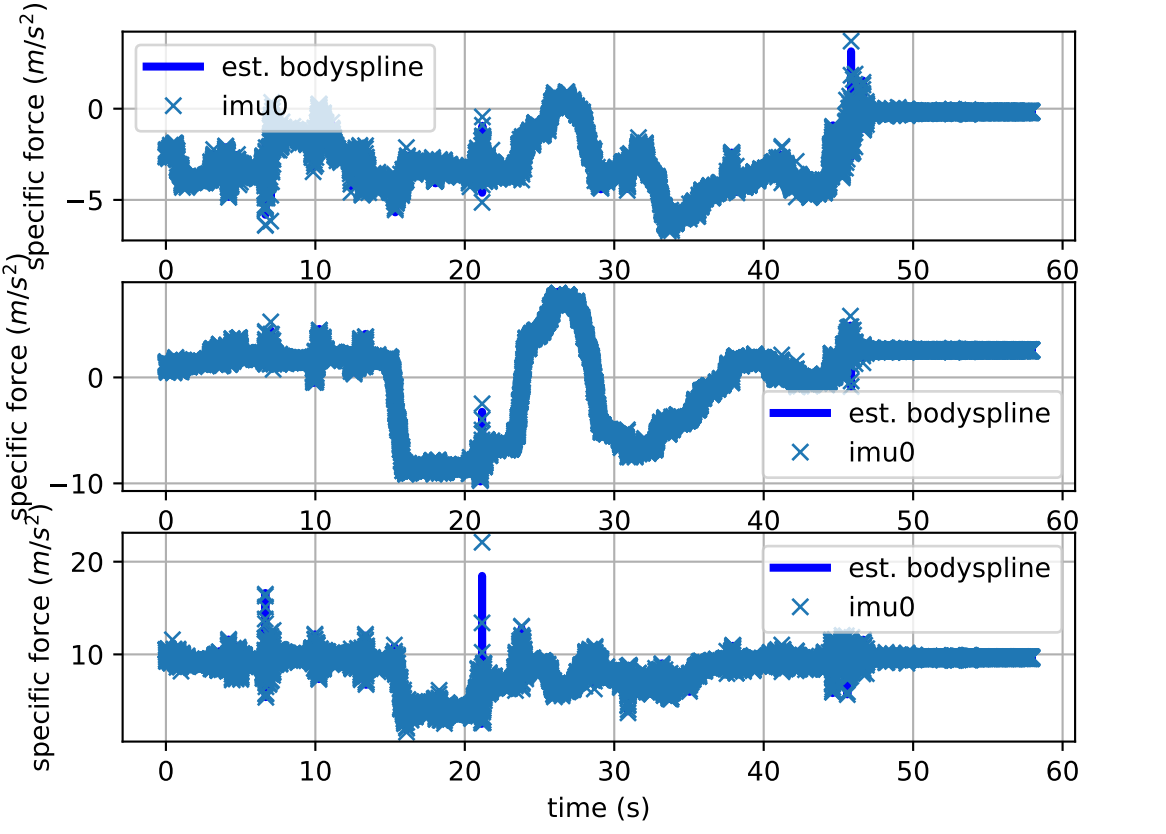
IMU0:

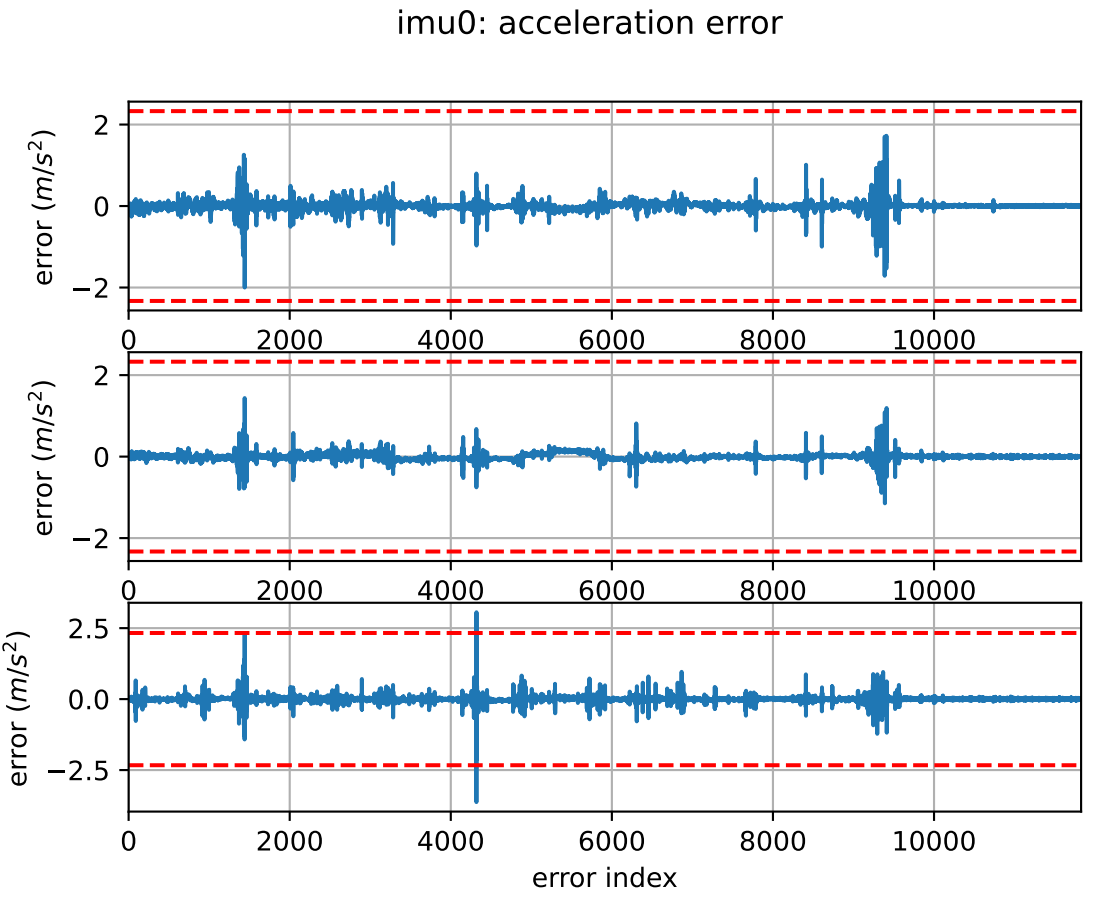
  Model: calibrated
  Update rate: 204.090470950253
  Accelerometer:
    Noise density: 0.054353532577651076
    Noise density (discrete): 0.7764958503851519
    Random walk: 0.013404724317075253
  Gyroscope:
    Noise density: 0.005527660738864506
    Noise density (discrete): 0.07896829189405931
    Random walk: 0.0002954052172196383

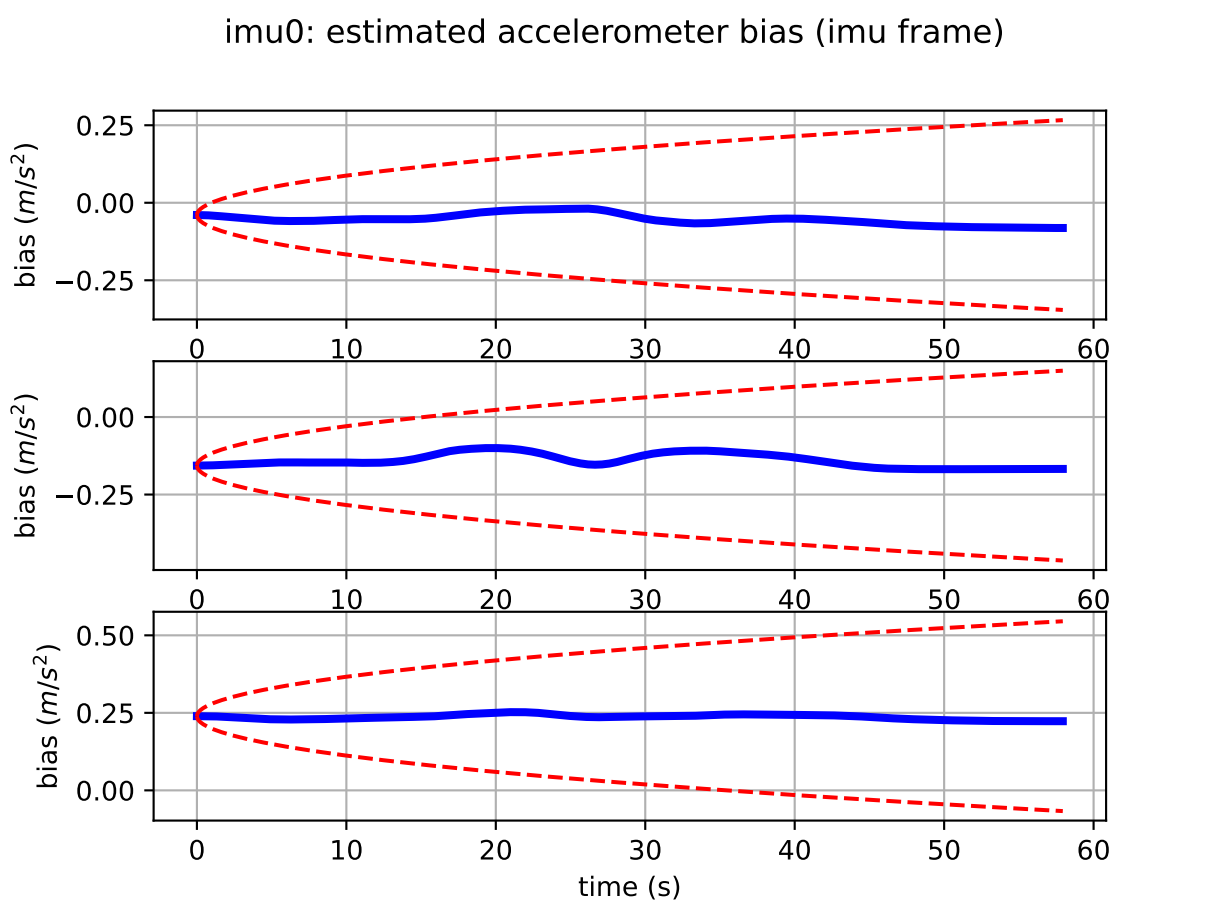
T_ib (imu0 to imu0)
  [[1. 0. 0. 0.]
   [0. 1. 0. 0.]
   [0. 0. 1. 0.]
   [0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]

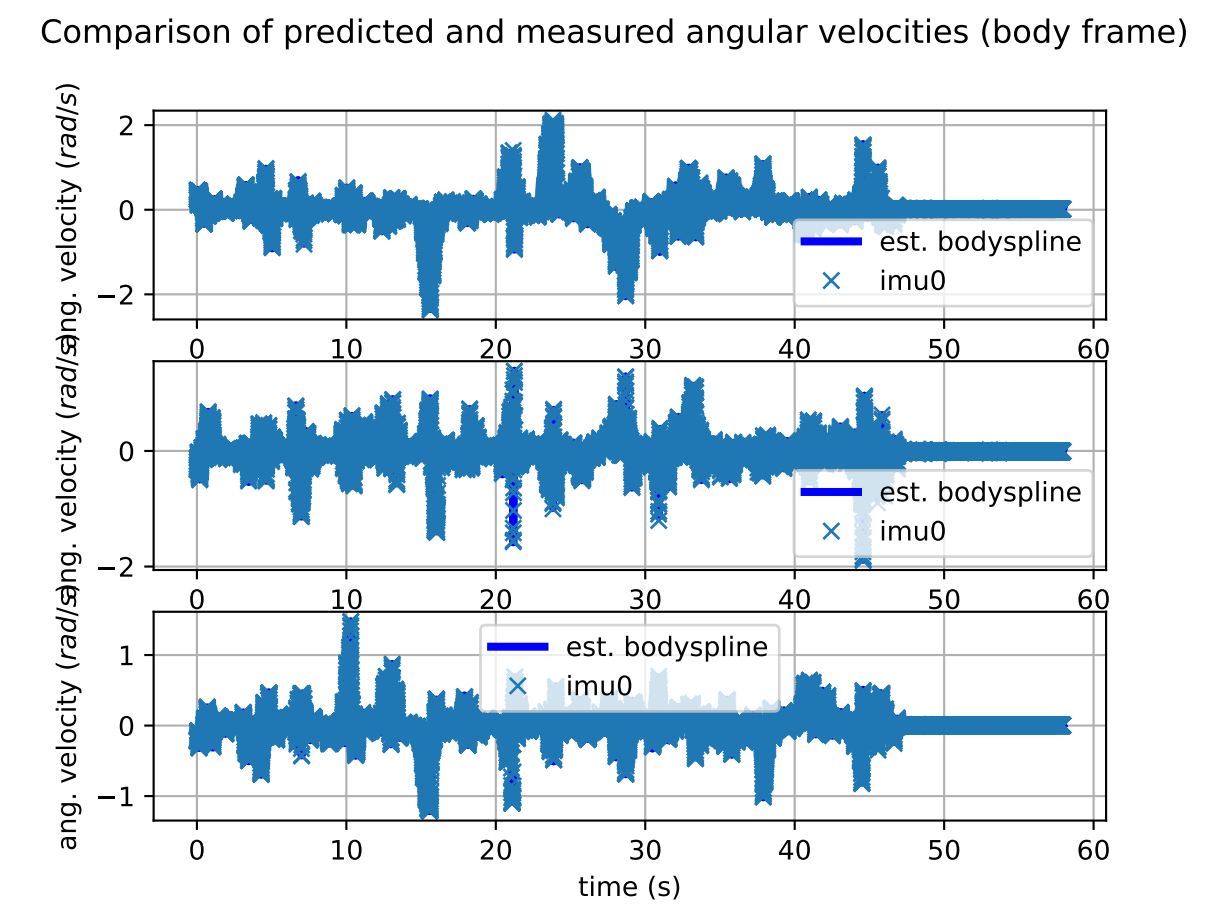

imu0: estimated poses
0.40
0.35
0.30
0.25
0.20
0.15
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−0.2
−0.1
0.0
0.1
0.2
0.3

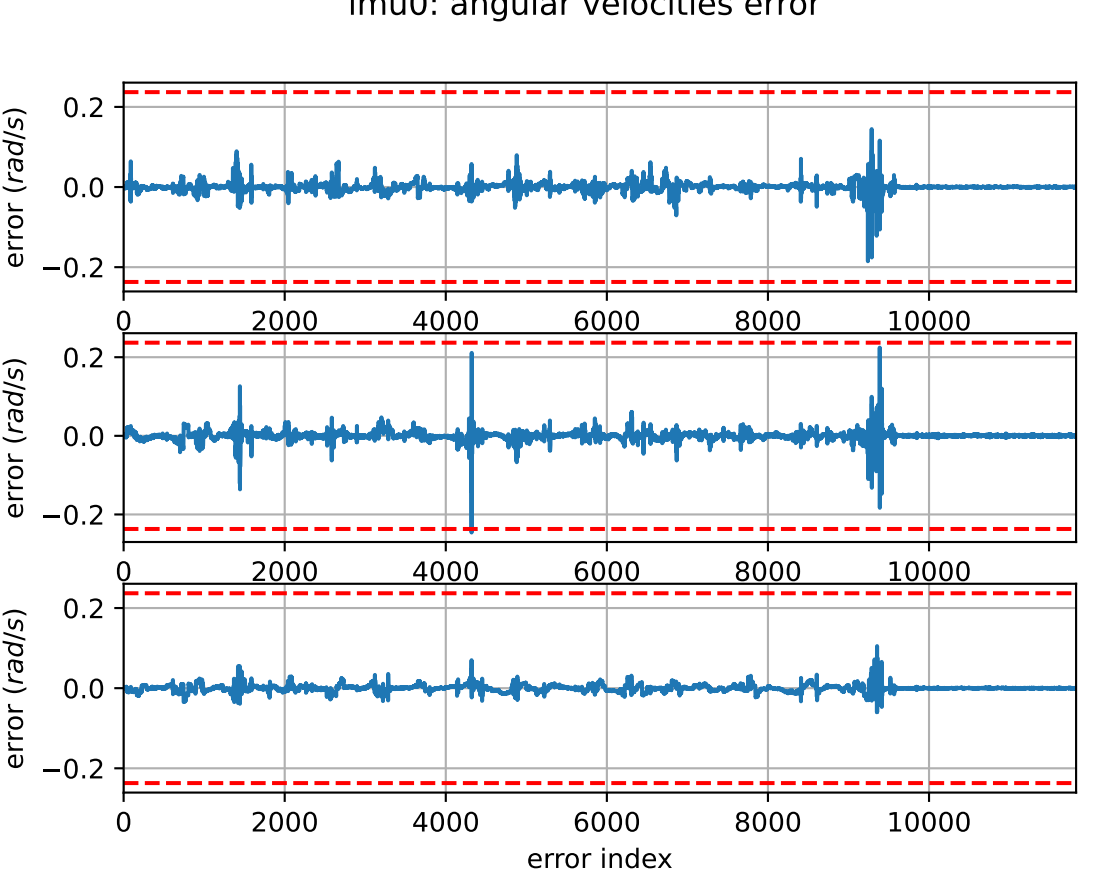

imu0: sample inertial rate
avg dt (ms) = 4.90 +- 0.0000
sample rate (ms)
0
1
2
3
4
10
20
30
40
50
time (s)

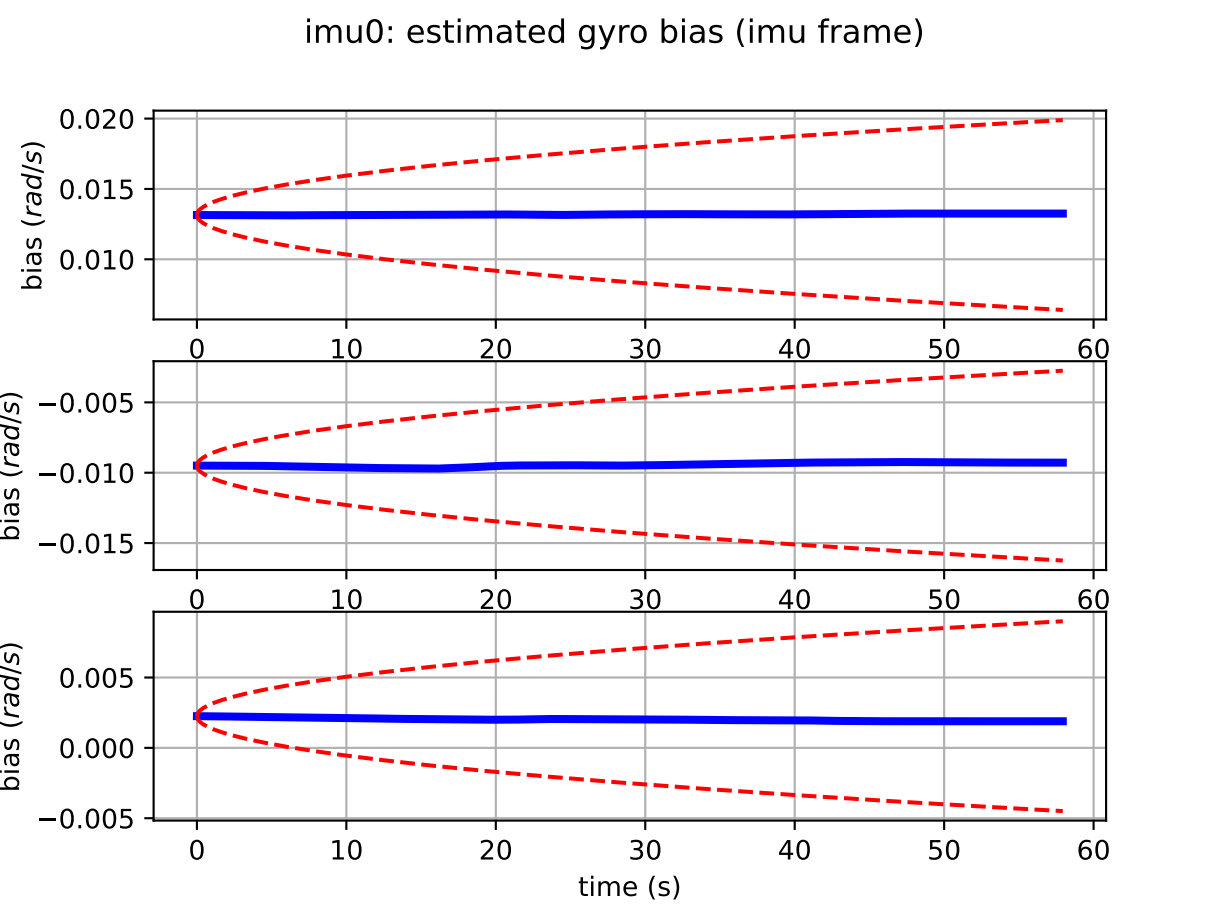

Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
est. bodyspline
imu0
0
−5
0
10
20
30
40
50
60
specific force (m/s^2)
est. bodyspline
imu0
0
−10
specific force (m/s^2)
est. bodyspline
imu0
20
10
time (s)

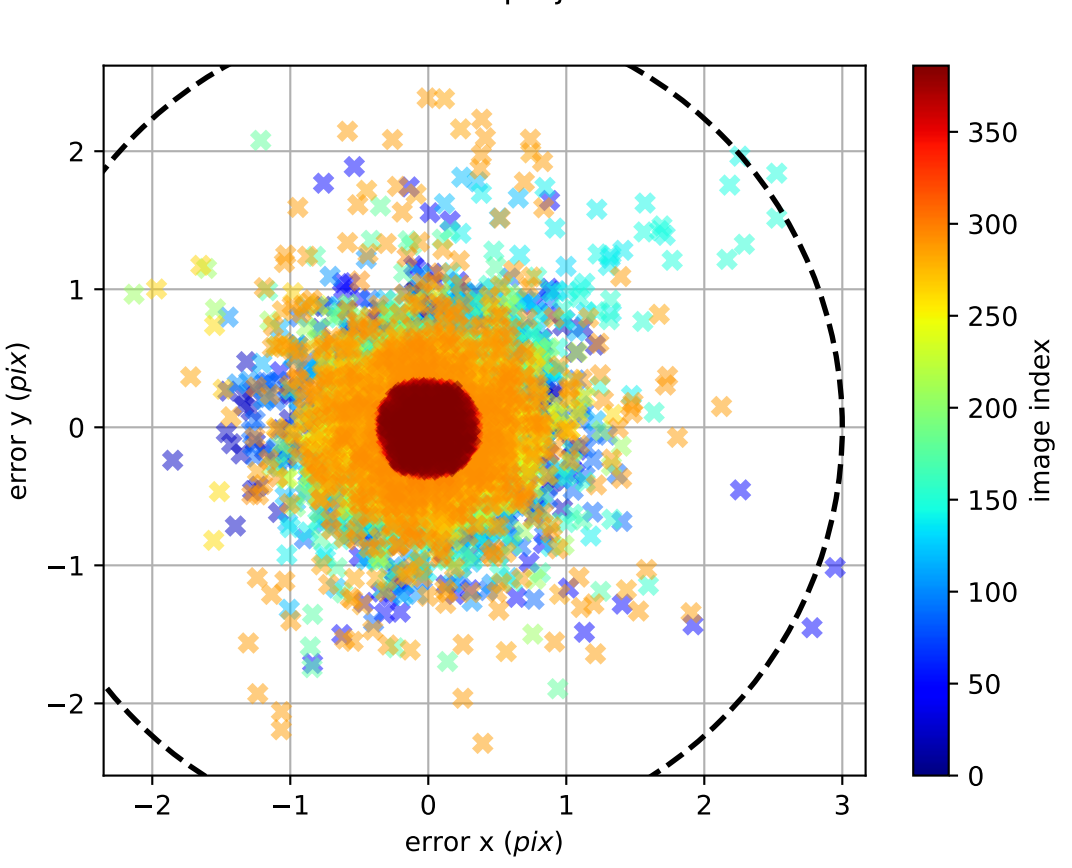

imu0: acceleration error
error (m/s^2)
2
0
−2
0
2000
4000
6000
8000
10000
error (m/s^2)
2
0
−2
0
2000
4000
6000
8000
10000
error (m/s^2)
2.5
0.0
−2.5
0
2000
4000
6000
8000
10000
error index


imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.25
0.00
−0.25
0
10
20
30
40
50
60
bias (m/s^2)
0.00
−0.25
0
10
20
30
40
50
60
bias (m/s^2)
0.50
0.25
0.00
0
10
20
30
40
50
60
time (s)


Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
ang. velocity (rad/s)
ang. velocity (rad/s)
time (s)
est. bodyspline
imu0
est. bodyspline
imu0
est. bodyspline
imu0


imu0: angular velocities error
error (rad/s)
0.2
0.0
−0.2
0
2000
4000
6000
8000
10000
error index


imu0: estimated gyro bias (imu frame)
bias (rad/s)
0.020
0.015
0.010
0
10
20
30
40
50
60
bias (rad/s)
−0.005
−0.010
−0.015
0
10
20
30
40
50
60
bias (rad/s)
0.005
0.000
−0.005
0
10
20
30
40
50
60
time (s)


cam0: reprojection errors
error x (pix)
error y (pix)
image index
−2
−1
0
1
2
3
2
1
0
−1
−2
350
300
250
200
150
100
50
0